# SIMATS ENGINEERING
## ASSESSMENT TOOL 1

**COURSE NAME: Artificial Intelligence** **COURSE CODE: CSA17**

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**COURSE OUTCOME COVERED**

***CO3:*** *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

*Assessment Tool Weightage: 50%*

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**Time Duration: 1 Hour**

**QUESTIONS: 4** **Total Marks:40**

Annexure A

## CASE STUDY

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***Code of Conduct:***

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

Annexure A

# CASE STUDY

**Case Study 1 — Smart Medical Diagnosis System**

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

I. If a patient has Fever AND Cough → Possible Flu

Ii. If a patient has Fever AND Rash → Possible Measles

Iii. If a patient has Cough AND Breathlessness → Possible Pneumonia

Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

**(a)** Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

**(b)** Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

**(c)** The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

**Case Study 2 — Autonomous Traffic Management Agent**

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

I. $\forall x$: Vehicle(x) $\wedge$ EmergencyType(x) $\rightarrow$ ClearPath(x) [Rule 1]

II. $\forall x$: ClearPath(x) $\rightarrow$ GreenSignal(x) $\wedge$ $\neg$RedSignal(x) [Rule 2]

III. $\forall x \forall y$: Vehicle(x) $\wedge$ Behind(x,y) $\wedge$ GreenSignal(y) $\rightarrow$ Proceed(x) [Rule 3]

IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

**(a)** Apply Unification to match Rule 1 with the given facts. Show the complete substitution set $\theta$ (MGU) at each step.

**(b)** Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

**(c)** Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

**Case Study 3 — Agricultural Expert Reasoning System**

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

1) P1: SoilDry → IrrigationNeeded
2) P2: IrrigationNeeded ∧ CropWheat → ApplyDripMethod
3) P3: ¬ApplyDripMethod ∧ CropWheat → CropAtRisk
4) Facts: SoilDry = True, CropWheat = True

**(a)** Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication → move ¬ inward → distribute).

**(b)** Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause □. Draw the resolution proof tree.

**(c)** If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

**Case Study 4 — Wumpus World Knowledge Agent**

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

1) Stench [1,2] → Wumpus is adjacent to cell [1,2]
2) Breeze [1,1] → Pit is adjacent to cell [1,1]
3) Glitter [x,y] → Gold is at [x,y]
4) ¬Wumpus [x,y] ∧ ¬Pit[x,y] → Safe[x,y]

**(a)** Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

**(b)** Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

**(c)** Evaluate whether the path [1,1] → [1,2] → [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

## RUBRICS FOR CALCULATION

| Criteria | Weigh tage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Understan ding of Case | 25 | Thorough understanding; clearly identifies all key issues | Good understanding with most issues identified | Basic understanding; some issues missed | Poor understanding ; problem not identified correctly |
| Application of Concepts | 25 | Effectively applies relevant concepts to analyze the case deeply | Applies appropriate concepts with minor gaps | Limited application of concepts | Fails to apply relevant concepts |
| Analysis | 20 | In-depth analysis with strong reasoning and insights | Good analysis with logical reasoning | Basic analysis with limited depth | Weak or no analysis; lacks reasoning |
| Solution Design | 20 | Innovative, feasible solutions with strong justification | Logical solutions with adequate justification | Basic solutions with weak justification | Incorrect or no solution provided |
| Clarity | 10 | Clear, well-structured, and easy to understand | Organized and understandable presentation | Limited clarity and structure | Poor clarity; difficult to understand |

# SIMATS ENGINEERING
# CO3–AT1: ARTIFICIAL INTELLIGENCE
# CASE STUDY ANSWERS

**Course Code: CSA17**

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Answers are based on the rules and facts in the supplied case study. Where the knowledge base is incomplete, the limitation is stated.

## CASE STUDY 1 — SMART MEDICAL DIAGNOSIS SYSTEM

### (a) Propositional Logic Representation

Symbols: F = Fever; C = Cough; R = Rash; B = Breathlessness; FL = Possible Flu; M = Possible Measles; P = Possible Pneumonia.

Rules: (1) $F \wedge C \rightarrow FL$. (2) $F \wedge R \rightarrow M$. (3) $C \wedge B \rightarrow P$.

Facts: F, C and B are true. Rash (R) is not given as a fact.

### (b) Forward Chaining

1. Initial facts: F, C, B.
2. Rule 1: $F \wedge C \rightarrow FL$. Both premises are true, so derive FL (Possible Flu).
3. Rule 2: $F \wedge R \rightarrow M$. R is not known to be true, so this rule cannot fire.
4. Rule 3: $C \wedge B \rightarrow P$. Both premises are true, so derive P (Possible Pneumonia).

Result: Possible Flu and Possible Pneumonia are derived. Possible Measles is not derived.

### (c) Backward Chaining for Pneumonia

Goal: P. Rule 3 reduces the goal to $C \wedge B$. C = True and B = True are known facts; therefore P is proved.

Goal tree:

P
└── $C \wedge B$
  ├── C (fact: True)
  └── B (fact: True)

Conclusion: The system derives possible pneumonia, not a confirmed clinical diagnosis.

## CASE STUDY 2 — AUTONOMOUS TRAFFIC MANAGEMENT AGENT

### (a) Unification and MGU

Rule 1: $\forall x\ [Vehicle(x) \wedge EmergencyType(x) \rightarrow ClearPath(x)]$. Facts include Vehicle(Ambulance) and EmergencyType(Ambulance).

5. Unify Vehicle(x) with Vehicle(Ambulance): $\theta_1 = \{x/Ambulance\}$.

6. Apply $\theta_1$ to EmergencyType(x), obtaining EmergencyType(Ambulance), which matches the given fact.

Most general unifier: $\theta = \{x/Ambulance\}$. The rule concludes ClearPath(Ambulance).

### (b) Resolution proof of Proceed(CarA)

CNF clauses:

- C1: ¬Vehicle(x) ∨ ¬EmergencyType(x) ∨ ClearPath(x)
- C2: ¬ClearPath(x) ∨ GreenSignal(x)
- C3: ¬ClearPath(x) ∨ ¬RedSignal(x)
- C4: ¬Vehicle(x) ∨ ¬Behind(x,y) ∨ ¬GreenSignal(y) ∨ Proceed(x)
- C5: Vehicle(Ambulance); C6: EmergencyType(Ambulance)
- C7: Vehicle(CarA); C8: Behind(CarA,Ambulance)

Add the negated goal C9: ¬Proceed(CarA). Resolution steps:

7. 1. C1 + C5, with x = Ambulance, gives ¬EmergencyType(Ambulance) ∨ ClearPath(Ambulance).
8. 2. Resolve with C6 to obtain ClearPath(Ambulance).
9. 3. Resolve C2 with ClearPath(Ambulance) to obtain GreenSignal(Ambulance).
10. 4. Resolve C4 with C7, x = CarA: ¬Behind(CarA,y) ∨ ¬GreenSignal(y) ∨ Proceed(CarA).
11. 5. Resolve with C8, y = Ambulance: ¬GreenSignal(Ambulance) ∨ Proceed(CarA).
12. 6. Resolve with GreenSignal(Ambulance) to obtain Proceed(CarA).
13. 7. Resolve Proceed(CarA) with C9 (¬Proceed(CarA)) to obtain the empty clause □.

Since resolution produces □, the negated goal is inconsistent with the KB; hence Proceed(CarA) is proved.

### (c) Two simultaneous emergency vehicles

The rules can process each emergency vehicle when its facts are present, but do not specify how to handle conflicting routes or simultaneous requests at one intersection.

- Add facts for each vehicle, its emergency type, destination, route and current intersection.
- Add rules to detect conflicting route requests and prevent incompatible green signals.
- Add a priority or queueing rule for competing emergencies, including a tie-breaking policy.
- Add facts/rules for road occupancy, route availability and intersection status.

Therefore, the current KB is insufficient for safe coordination of simultaneous emergency vehicles without conflict detection, priority, and intersection-state rules.

## CASE STUDY 3 — AGRICULTURAL EXPERT REASONING SYSTEM

### (a) CNF conversion

P1: SoilDry → IrrigationNeeded. Eliminate implication (A → B ≡ ¬A ∨ B): C1 = ¬SoilDry ∨ IrrigationNeeded.

P2: (IrrigationNeeded ∧ CropWheat) → ApplyDripMethod. Eliminate implication and apply De Morgan's law: C2 = ¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod.

P3: (¬ApplyDripMethod ∧ CropWheat) → CropAtRisk. Eliminate implication and apply De Morgan's law: C3 = ApplyDripMethod ∨ ¬CropWheat ∨ CropAtRisk.

Facts as CNF unit clauses: C4 = SoilDry; C5 = CropWheat. Each clause is a disjunction of literals (or a single literal), so the KB is in CNF.

### (b) Resolution refutation for ApplyDripMethod

Goal: ApplyDripMethod. Add its negation C6 = ¬ApplyDripMethod.

14. 1. Resolve C1 (¬SoilDry ∨ IrrigationNeeded) with C4 (SoilDry) to get IrrigationNeeded.
15. 2. Resolve C2 with IrrigationNeeded to get ¬CropWheat ∨ ApplyDripMethod.
16. 3. Resolve with C5 (CropWheat) to get ApplyDripMethod.
17. 4. Resolve ApplyDripMethod with C6 (¬ApplyDripMethod) to get □.

Proof tree:
SoilDry + C1 → IrrigationNeeded
IrrigationNeeded + CropWheat + C2 → ApplyDripMethod
ApplyDripMethod + ¬ApplyDripMethod → □

The empty clause proves ApplyDripMethod from the KB.

### (c) Removing SoilDry

18. Without the SoilDry fact, C1 cannot be used to derive IrrigationNeeded.
19. Without IrrigationNeeded, C2 cannot derive ApplyDripMethod from the remaining facts.
20. CropWheat remains true, but P3 requires both ¬ApplyDripMethod and CropWheat to derive CropAtRisk.
21. The KB does not establish ¬ApplyDripMethod. Failure to prove ApplyDripMethod does not mean its negation is true.

Conclusion: Neither ApplyDripMethod nor CropAtRisk follows from the remaining KB. CropAtRisk would follow if ¬ApplyDripMethod were separately established.

## CASE STUDY 4 — WUMPUS WORLD KNOWLEDGE AGENT

### (a) Belief state and Modus Ponens

Initial percepts: Stench(1,2), Breeze(1,1), Glitter(2,2).

- Stench(1,2) implies that the Wumpus is in a cell adjacent to [1,2].
- Breeze(1,1) implies that a pit is in a cell adjacent to [1,1].
- Glitter(2,2) implies Gold(2,2).

Derived: Gold is at [2,2]; at least one neighbour of [1,2] is a possible Wumpus location; at least one neighbour of [1,1] is a possible pit location. The rules do not identify the exact hazard cells.

Safety rule: ¬Wumpus(x,y) ∧ ¬Pit(x,y) → Safe(x,y). Since no negative hazard facts are supplied, no specific Safe(x,y) conclusion can be derived.

### (b) Forward chaining: possible locations

Assuming a standard grid with four-neighbour adjacency and excluding cells outside the grid:

Possible Wumpus cells from Stench at [1,2]: [1,1], [1,3], [2,2].

Possible pit cells from Breeze at [1,1]: [1,2], [2,1].

These are candidates, not confirmed locations. The source does not define grid boundaries or provide further percepts, so exact locations cannot be determined uniquely.

### (c) Safety of path [1,1] → [1,2] → [2,2]

- Breeze at [1,1] means [1,2] may contain a pit.
- Stench at [1,2] means [2,2] may contain the Wumpus.
- Glitter confirms gold at [2,2], but does not establish that the cell is free of hazards.
- No negative Wumpus/Pit facts are available to prove the cells safe using the stated safety rule.

Conclusion: The path is not proven safe by the supplied KB. This does not mean every cell is definitely dangerous; the knowledge is insufficient to guarantee safety. More percepts or hazard-exclusion facts are needed before choosing the route.

## OVERALL CONCLUSION

These cases apply propositional logic, forward and backward chaining, unification, resolution refutation, CNF conversion, and reasoning with incomplete information. A logical agent must distinguish proven facts from possible locations and conclusions that cannot be established from the available knowledge base.